

3 Lecture 3: Fundamental Solutions

Fundamental solution and how to use it to solve the Poisson equation.

C. Fundamental solutions.

3.1 Fundamental solution for the Laplacian in $\mathbb{R}^n$

See Evans, §2.2.1.

Radial (or spherically symmetric) solutions. Set

$$r = |x| = (x_1^2 + \cdots + x_n^2)^{1/2}, \quad \frac{\partial r}{\partial x_i} = \frac{x_i}{r}. \quad (3.1)$$

Suppose

$$u(x) = v(r) \quad \text{is radially symmetric.}$$

Then

$$u_{x_i} = v'(r) \frac{x_i}{r}, \quad (3.2)$$

$$u_{x_i x_i} = v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right). \quad (3.3)$$

Therefore,

$$\Delta u = v''(r) + \frac{n-1}{r} v'(r) = r^{1-n} (r^{n-1} v'(r))'. \quad (3.4)$$

Hence $\Delta u = 0$ for $r > 0$ if and only if

$$r^{n-1} v'(r) = \text{constant},$$

so

$$v(r) = \begin{cases} b_1 \log r + C_1, & n = 2, \\ \frac{b_2}{r^{n-2}} + C_2, & n \geq 3, \end{cases} \quad b_i, C_i \text{ constant.} \quad (3.5)$$

For $n = 1$, v is affine in r ; in particular, a nonconstant radial solution has the form $u(x) = \text{const. } |x|$ up to an additive constant.

Definition 3.1 (Fundamental solution of the Laplacian). Let

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}}, & n \geq 3, \end{cases} \quad (3.6)$$

where

$$\alpha(n) = |B_1| \quad (3.7)$$

is the volume of the unit ball $B_1 \subset \mathbb{R}^n$.

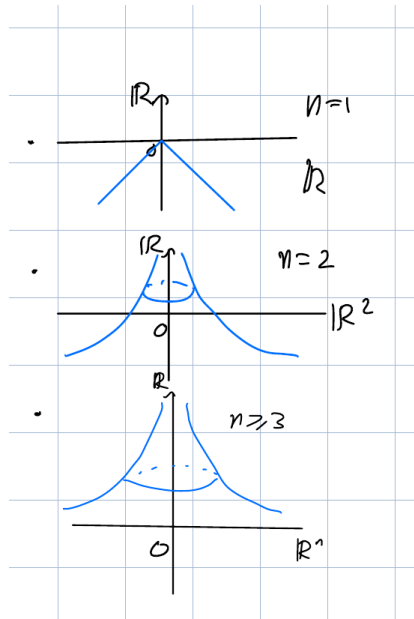


Figure 3.1: Original source sketches of radial profiles in dimensions $n = 1$, $n = 2$, and $n \geq 3$.

3.2 Singularity and local integrability

The fundamental solution Φ is singular at $x = 0$, while

$$\Delta\Phi(x) = 0, \quad x \neq 0 \text{ in } \mathbb{R}^n. \quad (3.8)$$

The source records the size estimates

$$|\Phi(x)| \sim \begin{cases} \text{const.} \cdot |\log |x||, & n = 2, \\ \frac{\text{const.}}{|x|^{n-2}}, & n \geq 3, \end{cases} \quad (3.9)$$

$$|D\Phi(x)| \sim \frac{\text{const.}}{|x|^{n-1}}, \quad (3.10)$$

$$|D^2\Phi(x)| \sim \frac{\text{const.}}{|x|^n}, \quad (3.11)$$

where the source notes in red that one may take $|D^2\Phi| = \max_{i,j} |D_{ij}\Phi|$.

For a bounded open set $U \subset \mathbb{R}^n$ with $0 \in U$, the source emphasizes:

$$\int_U |\Phi| \, dx < \infty \quad (\Phi \text{ is integrable near } 0), \quad (3.12)$$

$$\int_U |D\Phi| \, dx < \infty \quad (D\Phi \text{ is integrable near } 0), \quad (3.13)$$

$$\int_U |D^2\Phi| \, dx = \infty \quad (D^2\Phi \text{ is not integrable near } 0). \quad (3.14)$$

Question. Why are the coefficients $-1/(2\pi)$ and $1/[n(n-2)\alpha(n)]$ important? See below.

3.3 The fundamental-solution identity

Exercise 3.2 (Important fundamental-solution identity). Show that for every $\varphi \in C_c^2(\mathbb{R}^n)$,

$$-\int_{\mathbb{R}^n} (\Delta\varphi)(y) \Phi(y) dy = \varphi(0). \quad (3.15)$$

(Not easy.)

The source then notes, by translation, that

$$\varphi(x) = -\int_{\mathbb{R}^n} (\Delta\varphi)(y) \Phi(x-y) dy. \quad (3.16)$$

Hint: Green's formula. For $f, g \in C^2$ on a smooth domain Ω ,

$$\int_{\Omega} (\Delta f)g dx = \int_{\Omega} f\Delta g dx + \int_{\partial\Omega} \left(g \frac{\partial f}{\partial \nu} - f \frac{\partial g}{\partial \nu} \right) dS. \quad (3.17)$$

Apply Green's formula to

$$B_R \setminus B_{\varepsilon}(0), \quad \varepsilon \ll 1, \quad R \gg 1. \quad (3.18)$$

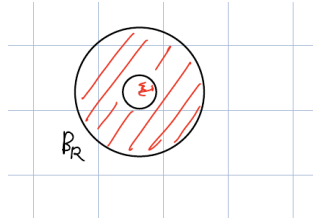


Figure 3.2: Original annular-domain sketch used in the Green's-formula hint.

Remark 3.3 (Formal integration by parts and the Dirac delta). If we formally integrate by parts in (3.15), then

$$\int_{\mathbb{R}^n} \varphi(y) [-\Delta\Phi(y)] dy = \varphi(0).$$

Thus one writes, *formally*,

$$-\Delta\Phi = \delta_0. \quad (3.19)$$

Definition 3.4 (Dirac delta). The Dirac delta at 0 is characterized by

$$\int_{\mathbb{R}^n} \delta_0(y) \varphi(y) dy = \varphi(0) \quad (3.20)$$

for every compactly supported continuous test function φ ; equivalently, in distribution notation,

$$\langle \delta_0, \varphi \rangle = \varphi(0).$$

It has unit mass concentrated at $x = 0$.

For a general point x , the source writes

$$-\Delta_y \Phi(y-x) = \delta_0(y-x) = \delta_x(y), \quad (3.21)$$

where δ_x denotes the Dirac mass at x . Equivalently,

$$\varphi(x) = -\int_{\mathbb{R}^n} \varphi(y) \Delta_y \Phi(y-x) dy \quad (3.22)$$

when interpreted distributionally.

3.4 Physical interpretation and superposition

Poisson equation.

$$-\Delta u = f, \quad u = \text{potential}, \quad f = \text{electric charge.} \quad (3.23)$$

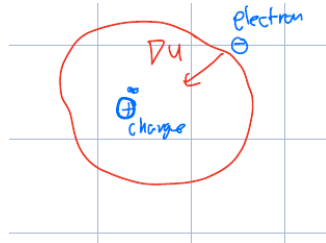


Figure 3.3: Original source sketch for the electrostatic interpretation of the potential.

With this sign convention, a positive electric charge corresponds to a local decrease of the potential away from the charge; the source summarizes this by writing $\Delta u = -$ “electric positive charge.”

The function $x \mapsto \Phi(x)$ represents the electric potential on $\mathbb{R}^n$ caused by a unit charge δ_0 at $x = 0$. More generally,

$$x \mapsto \Phi(x - y) \quad (3.24)$$

represents the electric potential caused by a unit charge δ_y at $x = y$.

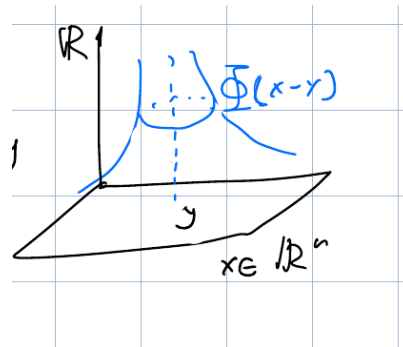


Figure 3.4: Original sketch of the translated fundamental solution $\Phi(x - y)$ centered at y .

Suppose electric charges are concentrated at $y_1, \dots, y_N$ with strengths $f_j, j = 1, \dots, N$. Then

$$f(x) = \sum_{j=1}^N f_j \delta_{y_j}(x) = \sum_{j=1}^N f_j \delta(x - y_j). \quad (3.25)$$

The corresponding electric potential is

$$u(x) = \sum_{j=1}^N \Phi(x - y_j) f_j, \quad (3.26)$$

by the superposition principle. Indeed,

$$-\Delta u = \sum_{j=1}^N [-\Delta \Phi(x - y_j)] f_j = \sum_{j=1}^N \delta(x - y_j) f_j = f. \quad (3.27)$$

Here, linearity of the operator Δ is crucially used. The superposition principle is a manifestation of linearity of the corresponding differential operator.

3.5 Continuous charge distributions and convolution

If the charge is distributed continuously with density $f(y)$ on $\mathbb{R}^n$, then the electric potential is

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy = (\Phi * f)(x). \quad (3.28)$$

Definition 3.5 (Convolution). For functions $f, g: \mathbb{R}^n \rightarrow \mathbb{R}$, define

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) \, dy, \quad (3.29)$$

whenever the integral is defined.

Exercise 3.6 (Basic properties of convolution). Whenever the relevant integrals converge absolutely, show that

- (i) $f * g = g * f$;
- (ii) $f * (g * h) = (f * g) * h$.

The fact that the integral is over all of $\mathbb{R}^n$ is important for these properties.